

ES-301 (GS)
B.Tech. I Semester
Examination, December 2025
Grading System (GS)
Energy and Environmental Engineering
Time: Three Hour

Maximum Marks: 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

- 1 (a) What is energy? Write various conventional and non-conventional energy sources.
- (b) What is solar energy? Discuss the importance and limitations of solar energy.
- 2 (a) Define ecology and ecosystem. Give a brief account of different components of an ecosystem.
- (b) What is food chain. Name and explain various types of food chains with suitable examples.
- 3 (a) What is Biodiversity? What are the major factors responsible for the loss genetic biodiversity? How can biodiversity be conserved?

- (b) Explain In-situ and Ex-situ conservation of biodiversity along with their merits and demerits.
- 4 (a) Explain Water pollution. Enlist various sources of water pollution giving their adverse effects. How water pollution can be minimized?
- (b) Explain the term solid waste. What are the adverse effects of solid waste in society? Discuss various methods for disposal of solid wastes.
- 5 (a) Explain the concept of Environmental Ethics. Discuss its importance in environmental decision-making.
- (b) Write short notes on:
 - (i) Resettlement and Rehabilitation (R&R)
 - (ii) Rainwater harvesting
- 6 (a) Define nuclear energy? State the advantages and disadvantages of using nuclear energy as a source of electricity.
- (b) Discuss the use of biomass energy as non-conventional source of energy. What are the advantages of biomass energy.
- 7 (a) Explain Disaster management? Explain the causes, effects and control measures of flood.

- (b) What are the main goals of forest conservation Act 1980?
-
- 8 (a) Explain about main causes, effects and control measures of urban and industrial wastes.
 - (b) What is sustainable development? Explain its objectives and significance in modern society.